

Hassan Ahmed

Software Engineer (AI/ML)

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Professional Summary

AI/ML Engineer with hands-on experience building RAG systems, Agentic AI workflows, NLP applications, and machine learning solutions using Python. Skilled in LangChain, Hugging Face, FastAPI, ChromaDB, Ollama, and Transformer-based models. Passionate about developing production-ready AI systems that combine LLMs, semantic search, and backend engineering to solve real-world problems.

Skills

- Technical Skills:** Python, OOP, SQL, Django, Flask, FastAPI, REST APIs, scikit-learn, TensorFlow, PyTorch, NLP, Transformers, Hugging Face, Pandas, NumPy, Matplotlib, Seaborn, Selenium, BeautifulSoup, LangChain, LangGraph, LLM, RAG, Ollama, ChromaDB, Prompt Engineering, Git, GitHub, Jupyter Notebook, Streamlit
- Soft Skills:** Problem Solving, Critical thinking, Communication, Networking, Presentation, Leadership, Teamwork

Projects

Movie Recommendation System

Machine Learning/NLP

- Built a Movie Recommendation System using Collaborative Filtering and Content-Based Filtering techniques to provide personalized movie suggestions based on user preferences and ratings data.
- Implemented data preprocessing, feature extraction, similarity computation, and model evaluation using Python and its Libraries.

AI CV Shortlister

Python, ML, DL, NLP

- Built an automated CV screening system that ranks candidates against any Job Description using semantic similarity and machine learning.
- Implemented cosine similarity using Sentence Transformers (all-MiniLM-L6-v2) for intelligent skill matching with tiered full/partial scoring.

AI Knowledge Retrieval Assistant (RAG)

Python, LangChain, Llama3, ChromaDB, Streamlit

- Engineered a local Retrieval-Augmented Generation (RAG) chatbot capable of answering questions from custom datasets.
- Orchestrated document chunking, semantic embedding, and vector storage using LangChain and ChromaDB.
- Built an interactive Streamlit UI powered by Ollama Llama3 to display AI-generated answers grounded in retrieved contextual sources.

MultiAgent AI Research System

Python, Langchain, Ollama Llama3, BeautifulSoup, Streamlit

- Built a multi-agent AI research tool utilizing specialized LangChain agents. A Search Agent extract top website links and Reader Agent scrapes data from those sources fed into a local Ollama Llama LLM chain to autonomously generate a comprehensive research report.
- Critic Chain evaluates the output, assigning a quality score with actionable feedback for improvement, before allowing the user to download the final report.

Experience

Trainee AI Engineer (Sapphire Consulting Services Pvt Ltd)

Karachi (04/2026 - Present)

- Built a production-ready CV Shortlisting Tool using Sentence Transformers (all-MiniLM-L6-v2) and cosine similarity, reducing recruiter screening time by ~60% for high-volume hiring pipelines.
- Developed an Anomaly Detection system using Isolation Forest on structured operational data, achieving 90+% precision in flagging outliers across multi-dimensional feature sets.
- Engineered end-to-end data pipelines combining OCR, NLP preprocessing, and ML inference; integrated Computer Vision modules for document understanding tasks.

Education

B.E Software Engineering Mehran University of Engineering and Technology Jamshoro CGPA: 3.2/4.0 2021-2025